

NuMicro[®] Family VS Code Quick Start Guide

The information described in this document is the exclusive intellectual property of Nuvoton Technology Corporation and shall not be reproduced without permission from Nuvoton.

Nuvoton is providing this document only for reference purposes of NuMicro microcontroller based system design. Nuvoton assumes no responsibility for errors or omissions.

All data and specifications are subject to change without notice.

For additional information or questions, please contact: Nuvoton Technology Corporation.

www.nuvoton.com

TABLE OF CONTENTS

1 INSTALLATION VS CODE AND EXTENSIONS 3

2 GET STARTED WITH A SAMPLE PROJECT 4

3 MANAGE ARM LICENSE 7

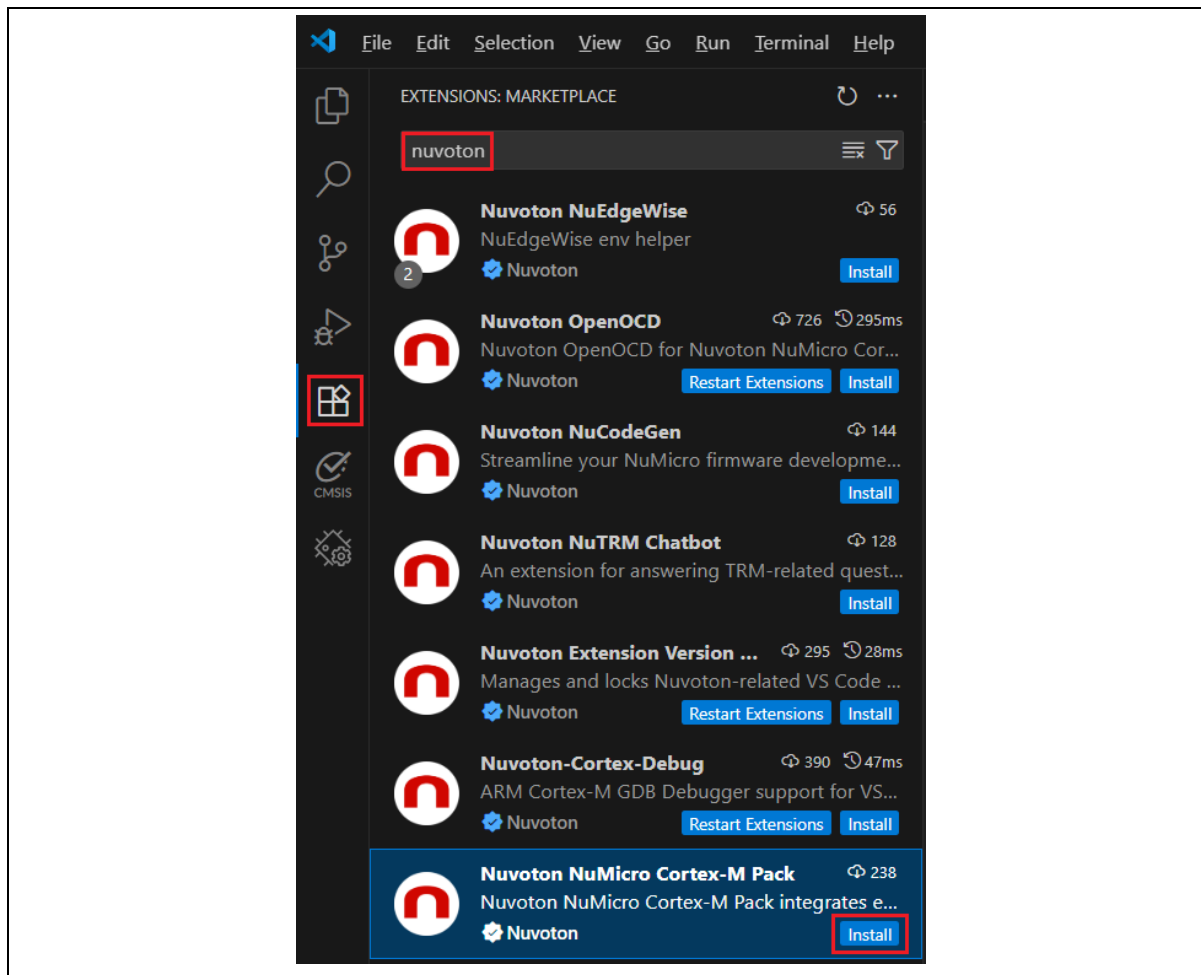
4 CONFIGURE THE DEVICE 9

5 RUN THE SAMPLE PROJECT 12

6 ADDITIONAL NOTE 15

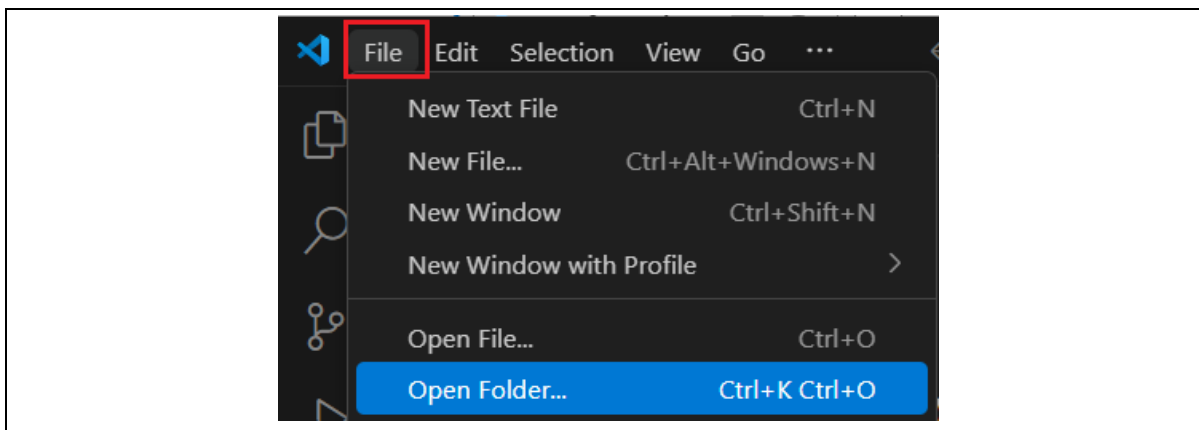
1 INSTALLATION VS CODE AND EXTENSIONS

1. Download VS Code from <https://code.visualstudio.com> and install it.
2. Launch VS Code and click Extensions in the Activity Bar.
3. Text “**Nuvoton NuMicro Cortex-M Pack Extension**” in search bar. Click the “**Install**” button to install.

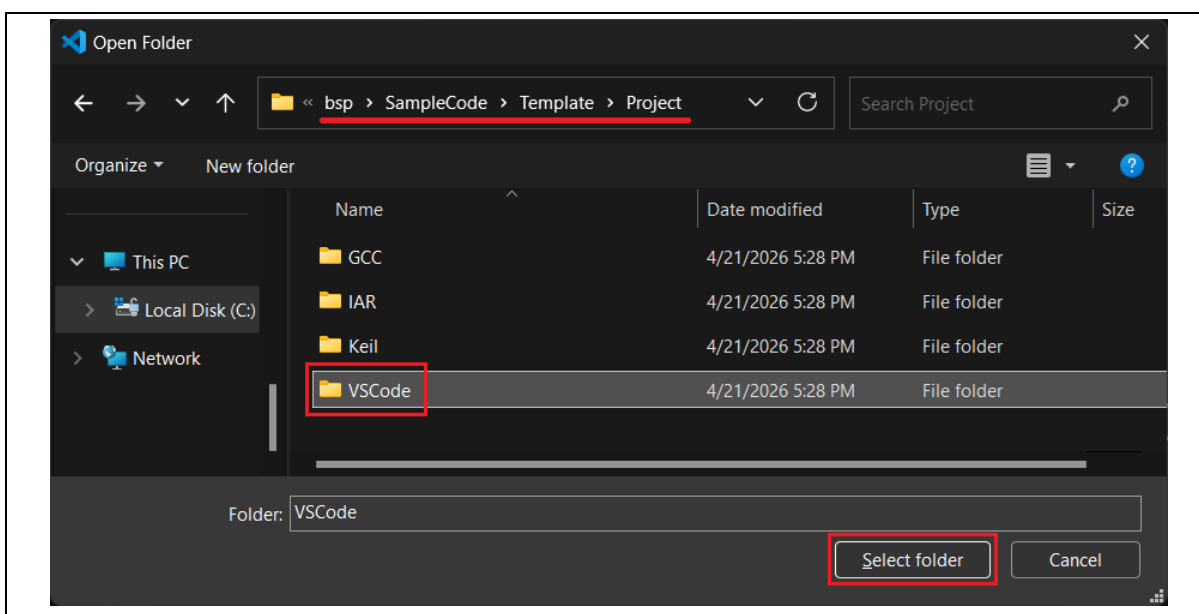


2 GET STARTED WITH A SAMPLE PROJECT

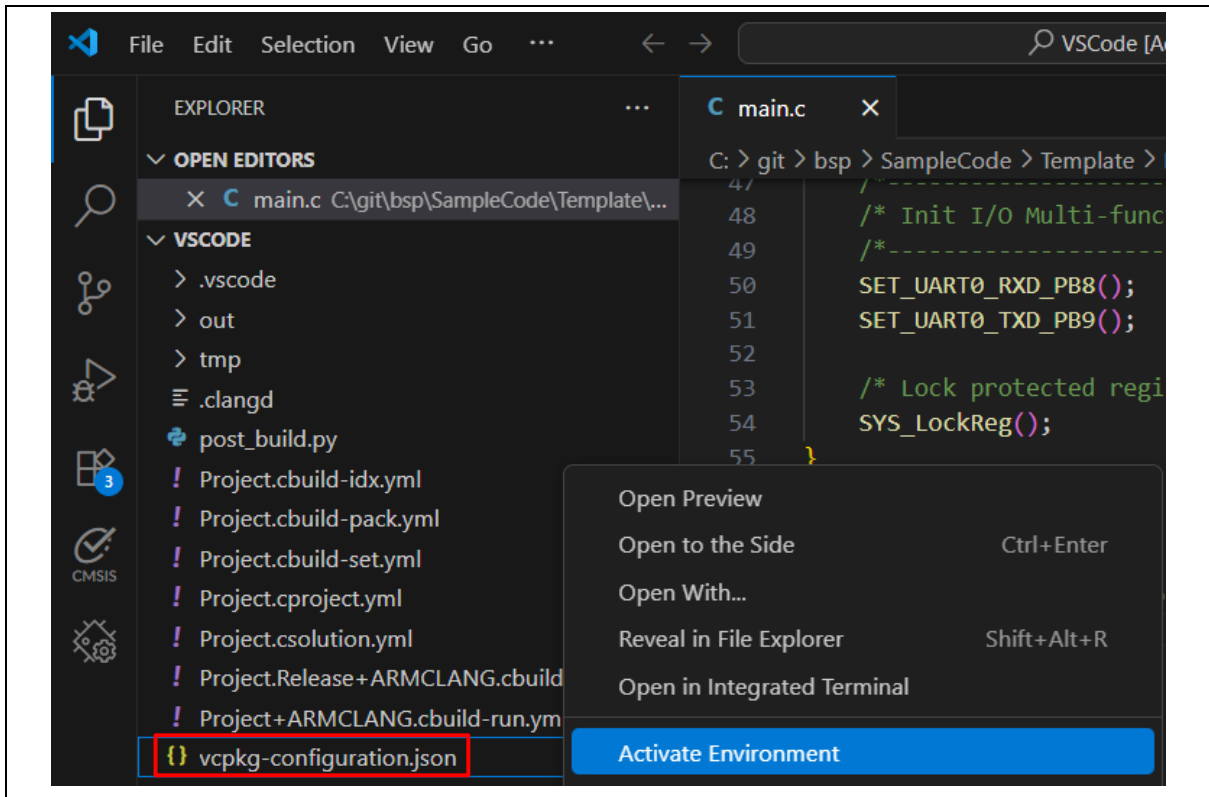
1. Click "File" menu and select "Open Folder" in the toolbar.



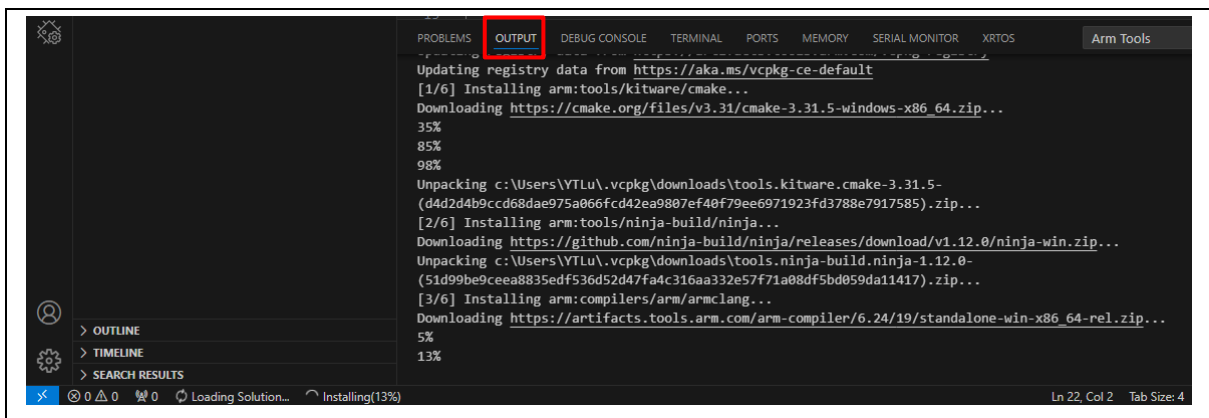
2. Select the "VSCode" folder of a sample project to open the project.



- Right-click “**vcpkg-configuration.json**” file and select “**Activate Environment**” command. (Please use private network connection.)

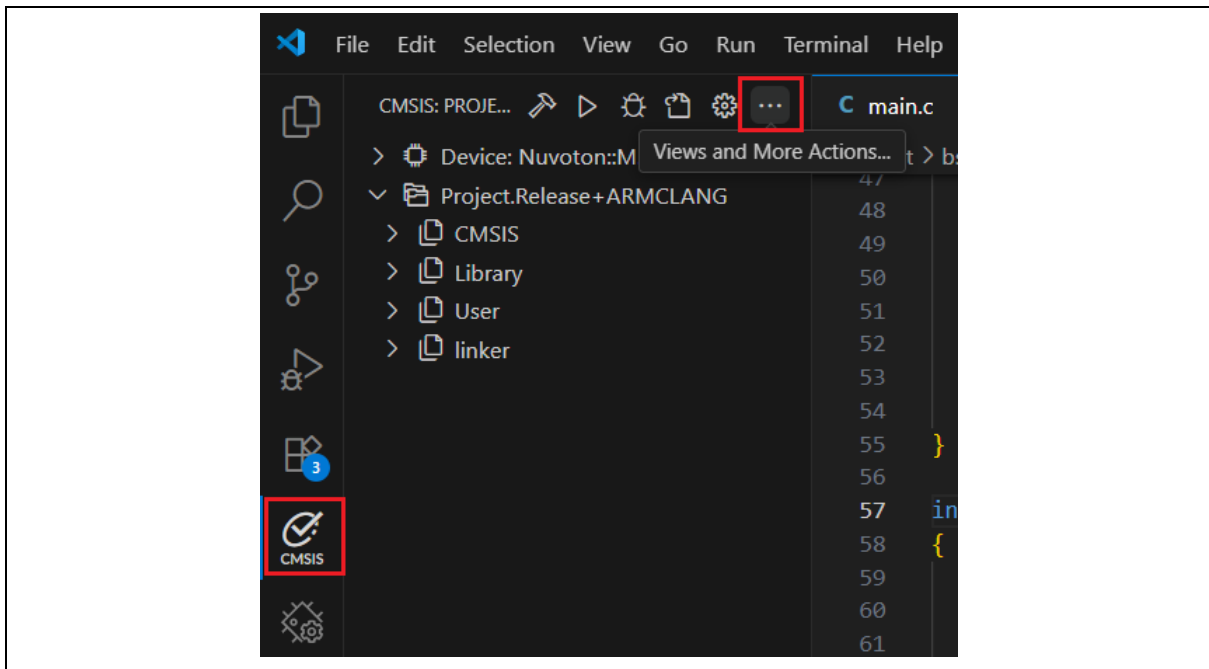


- Check **OUTPUT** terminal at the bottom. It will download and install the requires tools.

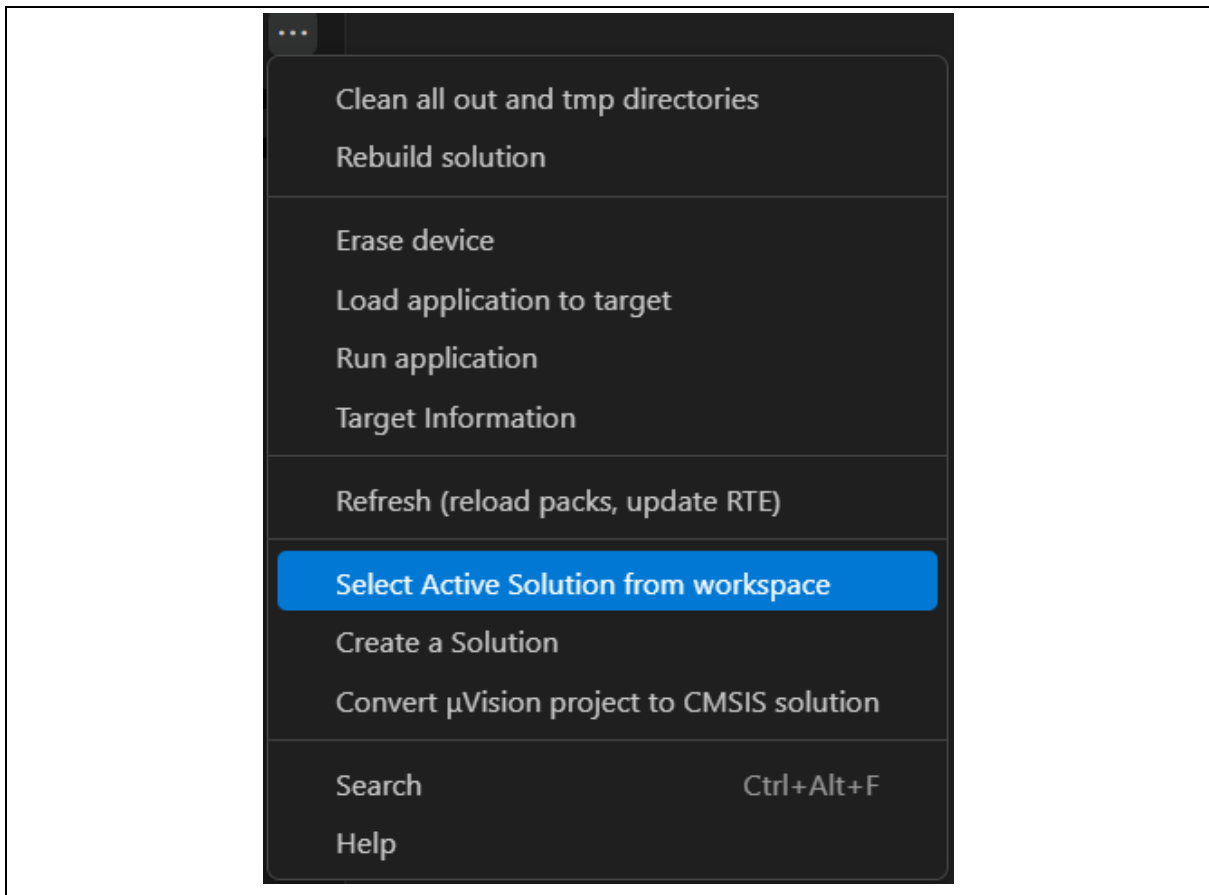


5. Click **CMSIS** icon in the Activity Bar.

Then click **“Views and More Commands”** button.

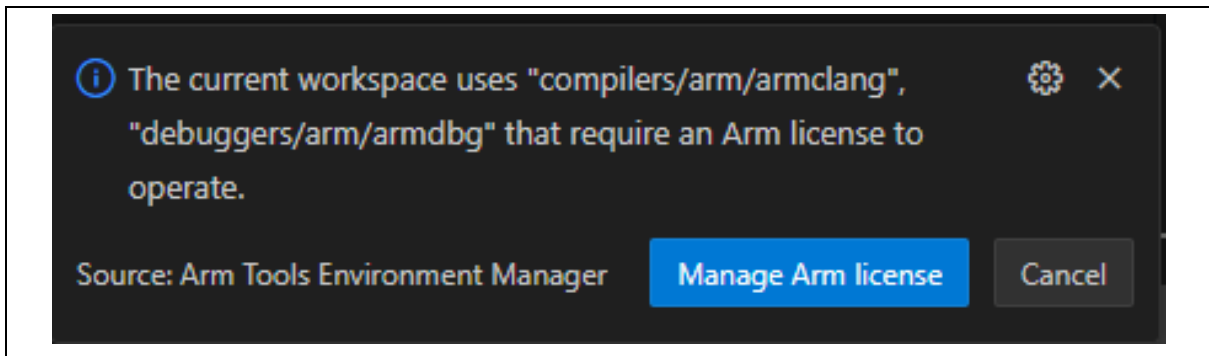


6. Select **“Select Active solution from workspace”** command.

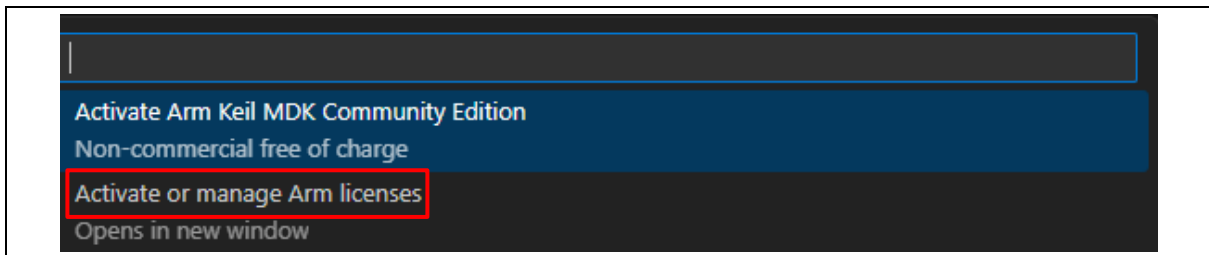


3 MANAGE ARM LICENSE

1. Click **"Manage Arm License"** button in the pop-up window at the bottom right.

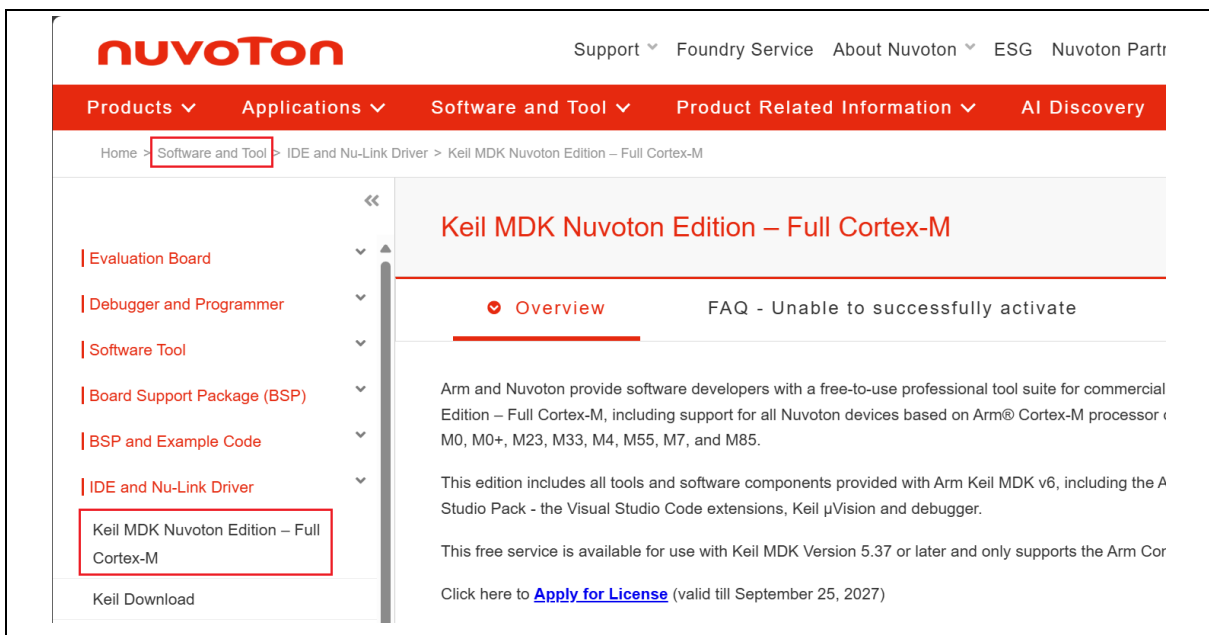


2. In the search bar at the top, select **"Activate or manage Arm licenses"** command from the available options.



3. Get Keil MDK License ID code.

- 1) Navigation to [Official Website](#).



2) Fill out the form.

Apply for Keil MDK Nuvoton Edition – Full Cortex-M

First Name*	Email*
Last Name*	Phone*
Company / Organization *	Industry
	-- Industry --
Job Title *	Application*
	-- Application --
Region / Country or region *	Series*
-- Select Region --	-- Select Line --
State / Province*	-- Select Category --
	Part No

NUVOTON Privacy Policy
 NUVOTON Privacy Policy
 Last modified: June 01, 2023

NUVOTON TECHNOLOGY CORPORATION and its affiliated companies (hereinafter referred to as "NUVOTON" or "us") are committed to protecting and respecting your privacy. Your privacy is important to us. We endeavor to comply with all applicable laws on privacy protection and personal information security. NUVOTON Privacy Policy, together with any additional terms and conditions that contain supplementary information in connection with particular services you are using (hereinafter referred to as "Privacy Policy"), outline our privacy practices regarding the collection, use and safeguard of your personal information through (1) the website(s) operated by NUVOTON (http://www.nuvoton.com) (the "Website(s)"), (2) NUVOTON's App, and (3) any services, both online and offline we provide (hereinafter referred to as "SERVICE"), along with whom we may share or disclose the collected information. If you are a minor, you shall access or use the SERVICE only after your parents (or your guardian) read and agree our Privacy Policy.

The use, access or submission of personal information to the SERVICE by you, or if you use a minor, the submission of your personal information either by your parents (or your guardian) or by you with the permission of your parents (or guardian) to NUVOTON will be deemed as your acceptance of this Privacy Policy.

☐ I have read and I accept Nuvoton's Privacy Policy

☐ Add me as my Nuvoton member

*If Nuvoton detects any suspicious behavior that violates Nuvoton Terms of Use, Privacy Policy, or related regulations on Nuvoton website (including other site regulations and relevant legal provisions), or if you engage in any illegal activities using the service or the Keil MDK Nuvoton Edition – Full Cortex-M, or if other abnormal situations arise (such as unauthorized use of your account by a third party), Nuvoton Technology has the right to suspend or permanently terminate your account and/or suspend or permanently terminate the related services and licenses. By clicking submit, you agree to this statement.

Submit

Let's chat
 +86 20 2210 2202
 chat@nuvoton.com

3) Check mailbox and fill in the License Server.

Arm License Management Utility

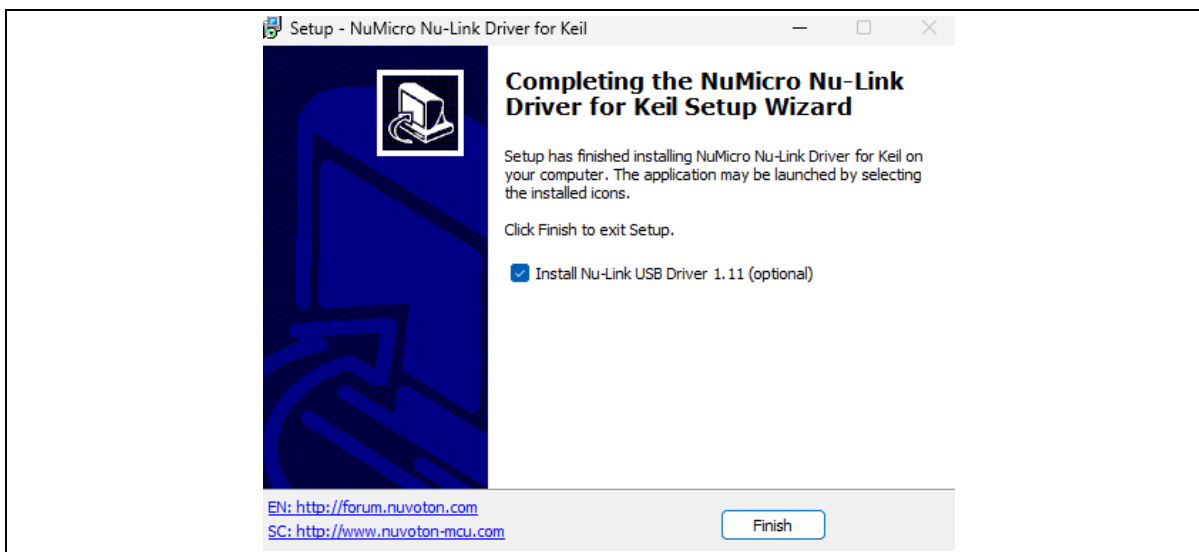
Activate with: ☐ Activation Code ☒ License Server

Enter your license server address:

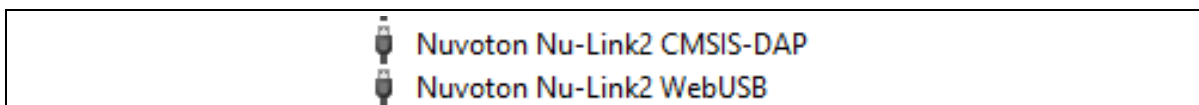
Query

4 CONFIGURE THE DEVICE

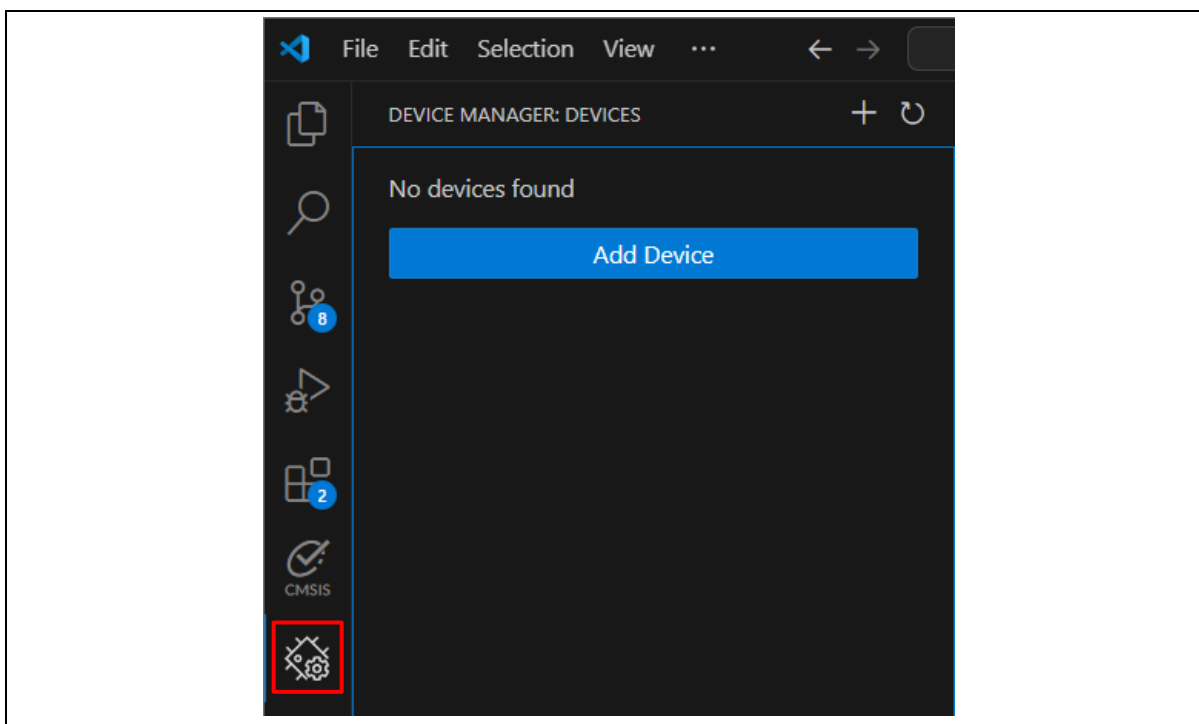
1. Install Nuvoton Nu-Link Keil Driver.
2. After installing the Keil driver, please check the box to install the Nu-Link USB Driver.



3. Installation complete in Windows Device Manager.



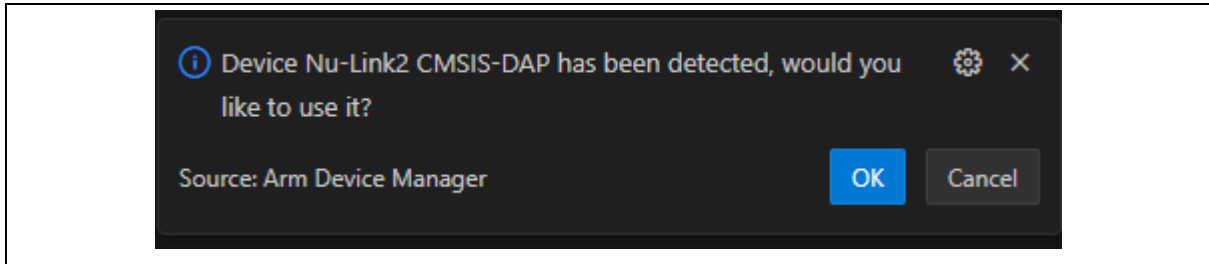
4. Click “**Device Manager**” icon in the Activity Bar to open the Device Manager.



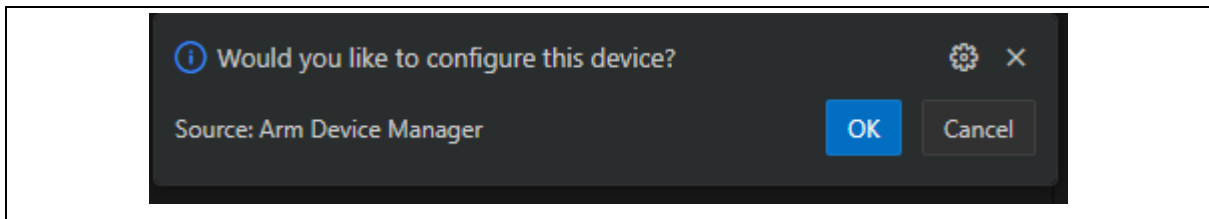
5. Connect device to your computer over USB.

The Device Manager detects the board and displays a pop-up message.

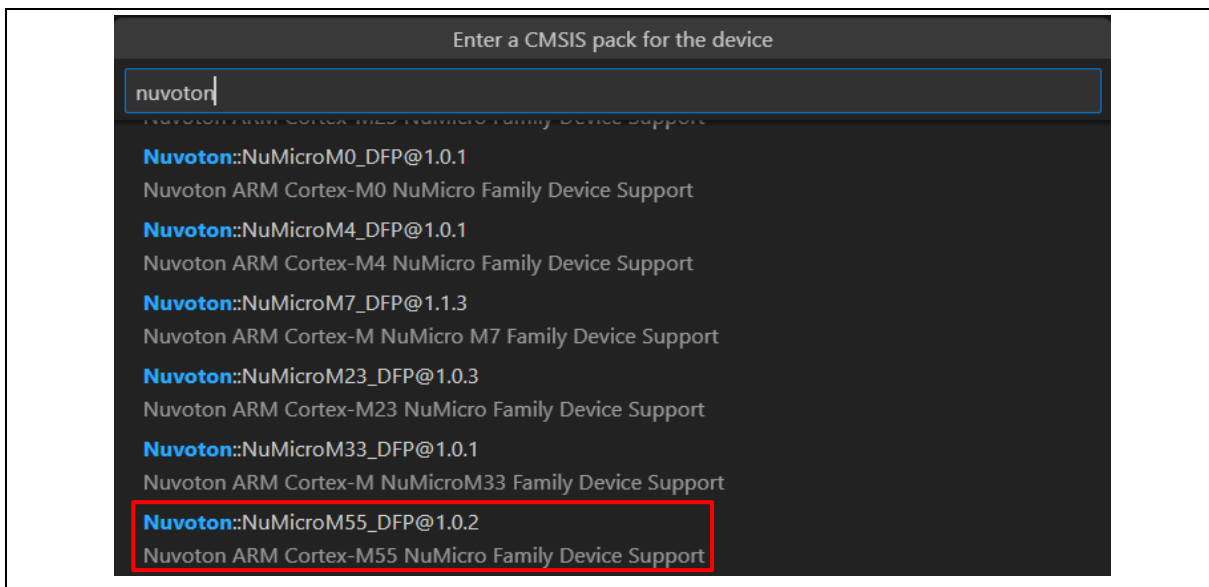
Press OK in the pop-up message and use it.



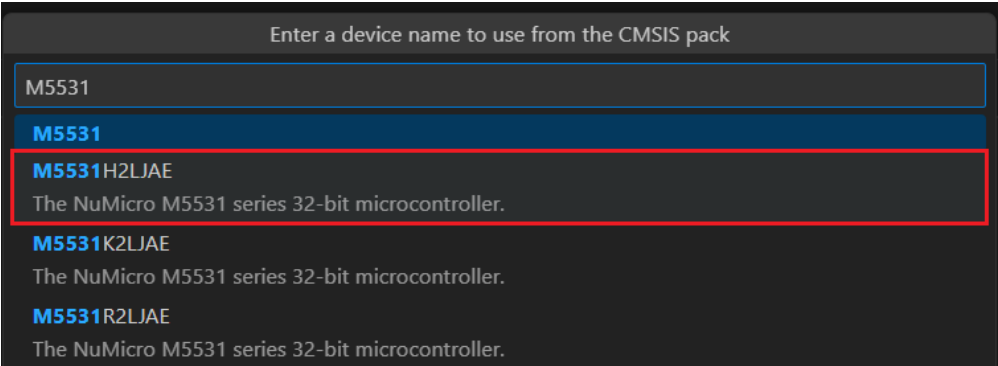
6. Press OK in the pop-up message and configure this device.



7. Text “**nuvoton**” in search bar and select corresponding CMSIS pack for the device.



8. Text your “**device model number**” in search bar.



Enter a device name to use from the CMSIS pack

M5531

M5531

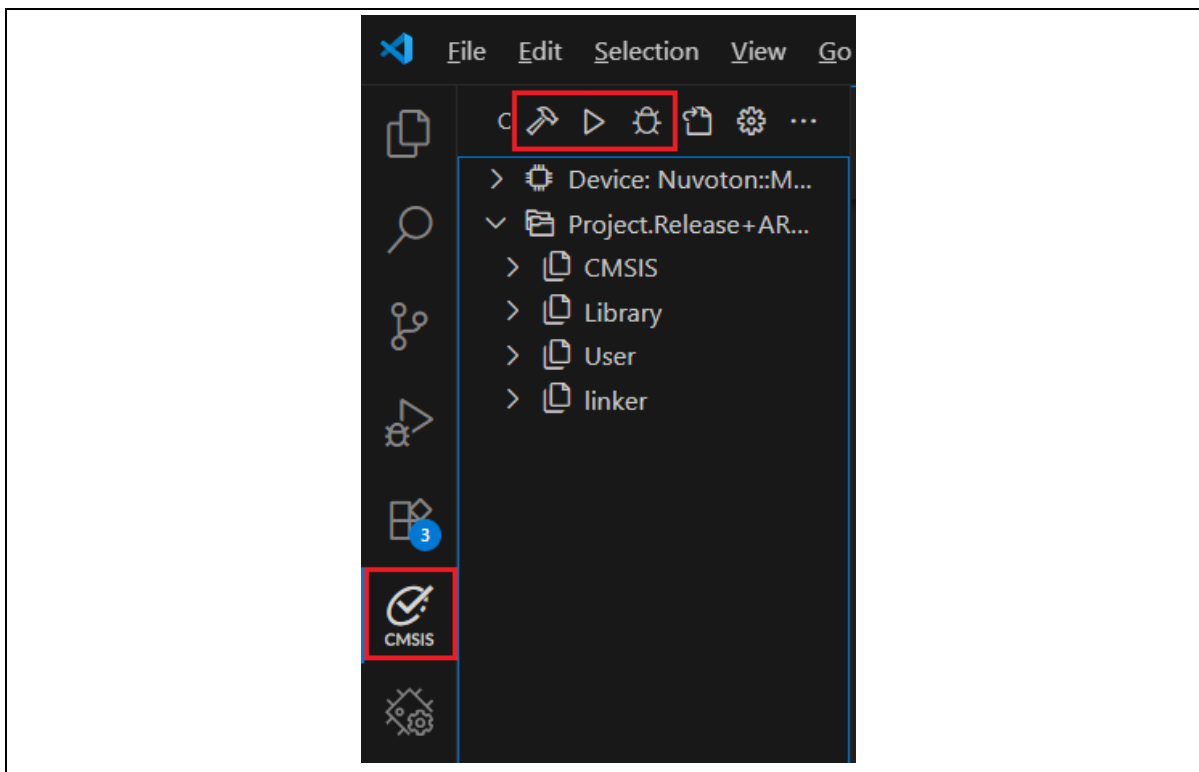
M5531H2LJAE
The NuMicro M5531 series 32-bit microcontroller.

M5531K2LJAE
The NuMicro M5531 series 32-bit microcontroller.

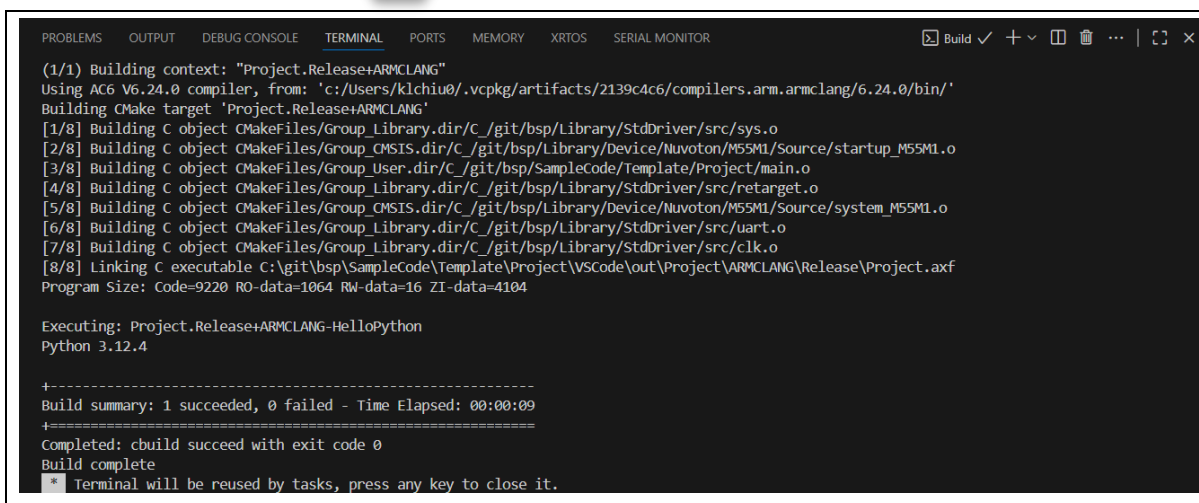
M5531R2LJAE
The NuMicro M5531 series 32-bit microcontroller.

5 RUN THE SAMPLE PROJECT

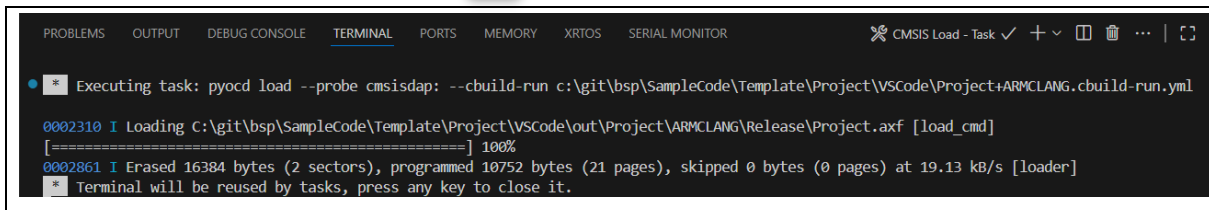
1. Click “CMSIS” icon in the Activity Bar.



2. Click “Build” button.

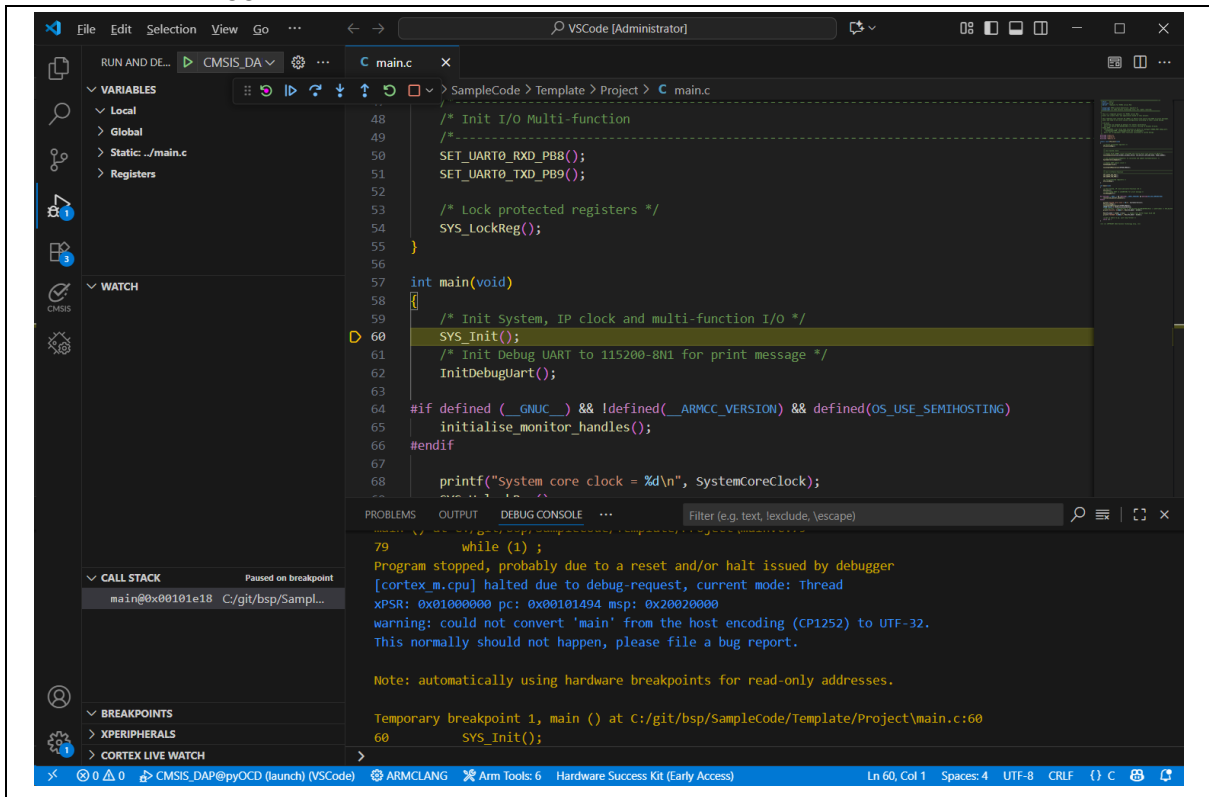


3. Click “Load & Run” button. 

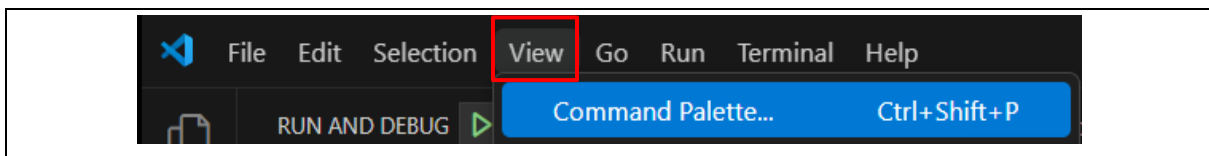


4. Click “Load & Debug” button. 

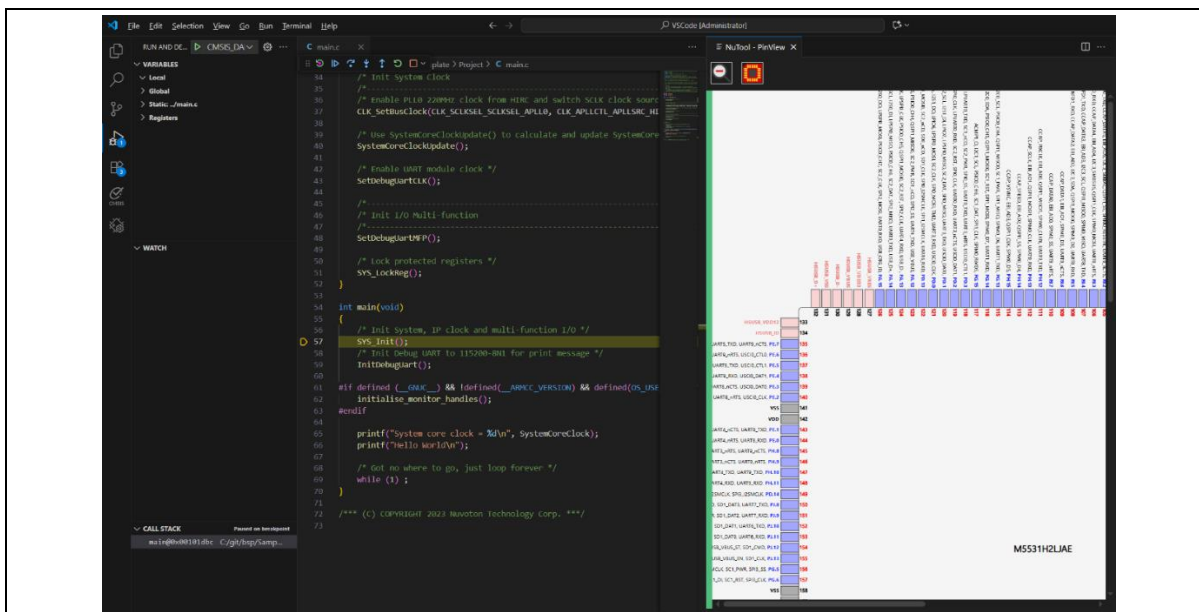
The debugger stops at the main function.



5. Click “View” menu and select “Command palette” item in the toolbar.



6. Text “Nuvoton:Open PinView” in search bar to open PinView tool.



6 ADDITIONAL NOTE

1. How to Fix Windows Long Path Errors in VS Code

If you encounter the following compilation errors while working in VS Code, your project likely exceeds the default Windows path limit:

- 1) The maximum full path to an object file is 250 characters.

```
ase/CMakeFiles/CMakeScratch/TryCompile-n0omuq/CMakeFiles/cmTC_a82c8.dir/./
has 251 characters. The maximum full path to an object file is 250
characters (see CMAKE_OBJECT_PATH_MAX). Object file
CMakeCCompilerABI.o
cannot be safely placed under this directory. The build may not work
correctly.
```

- 2) The system cannot find the path specified.

```
LANG-stamp/Template.Release+ARMCLANG-configure-.cmake:49 (message):
Command failed: The system cannot find the path specified
```

To resolve this, follow these steps to enable Long Paths:

- 1) Open Registry Editor: Press Win + R, type regedit, and press Enter.
- 2) Navigate to: HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\FileSystem
- 3) Enable Feature: Double-click LongPathsEnabled and change the Value data from 0 to 1.
- 4) Apply: Click OK and restart your computer (or restart VS Code) to apply the changes.

For more technical details on why this limit exists and how Windows handles it, please refer to the official Microsoft documentation:

[Microsoft Learn: Maximum Path Length Limitation](#)

REVISION HISTORY

Date	Revision	Description
2026.02.12	1.01	1. Add additional note.
2025.10.29	1.00	1. Initial version.

Important Notice

Nuvoton Products are neither intended nor warranted for usage in systems or equipment, any malfunction or failure of which may cause loss of human life, bodily injury or severe property damage. Such applications are deemed, "Insecure Usage".

Insecure usage includes, but is not limited to: equipment for surgical implementation, atomic energy control instruments, airplane or spaceship instruments, the control or operation of dynamic, brake or safety systems designed for vehicular use, traffic signal instruments, all types of safety devices, and other applications intended to support or sustain life.

All Insecure Usage shall be made at customer's risk, and in the event that third parties lay claims to Nuvoton as a result of customer's Insecure Usage, customer shall indemnify the damages and liabilities thus incurred by Nuvoton.

*Please note that all data and specifications are subject to change without notice.
All the trademarks of products and companies mentioned in this datasheet belong to their respective owners.*